CHA-3.B.1 The derivative can be used to solve rectilinear motion problems involving position, speed, velocity, and acceleration.

4-2 Straight-Line Motion

One important application of derivatives in calculus is particle motion. The three equations that are important when working with particle motion are: $f(t)$ $x(t)$

1. **position** $s(t)$ determines where the particle is located on the x-axis at a given time t
2. **velocity** $v(t) = s'(t)$ the rate of change of position—determines how fast the position is changing at time *t*, as well as the direction of movement
3. **acceleration** $a(t) = v'(t) = s''(t)$ the rate of change of velocity—determines how fast the velocity is changing at time *t*; the sign indicates if the velocity is increasing or decreasing

Important Note: In Calculus AB, all motion is considered "rectilinear"—meaning motion only happens along a straight path (horizontally or vertically).

Speed vs. Velocity	Speed and velocity are NOT the same! **Speed** = $\lvert \boldsymbol{v}(\boldsymbol{t}) \rvert$ In physics, we say that speed is a scalar and velocity is a vector (it has direction)

Questions involving motion are common on the AP Exam. We will need to be able to use these equations and take derivatives as needed to answer questions involving motion. To answer questions on particle motion, we will need to be able to decode questions into math equations. Consider the following phrases…

Statement	Translation	Steps to Solve
… initially	$t = 0$	– – – – – – – –
Particle is at the origin	POS. $s(t) = 0$	– – – – – – – –
Displacement of the particle (over a time interval $[a, b]$)	Net change in position over the time interval	$s(b) - s(a)$
Particle is at rest… (or stopped)	$v(t) = 0$	Set $v(t) = 0$ and solve for t
Particle is moving to the right…	$v(t) > 0$	Set $v(t) = 0$ and use a sign chart or graph
Particle is moving to the left…	$v(t) < 0$	Set $v(t) = 0$ and use a sign chart or graph
Particle turns around…	$v(t)$ changes signs	Set $v(t) = 0$ and use a sign chart or graph
Particle is speeding up… (aka speed is increasing)	$v(t)$ and $a(t)$ have the same sign	Check the signs of $v(t)$ and $a(t)$ to see if they have the same sign
Particle is slowing down… (aka speed is decreasing)	$v(t)$ and $a(t)$ have opposite signs	Check the signs of $v(t)$ and $a(t)$ to see if they have different signs

Example 1: For $0 \le t \le 8$, the position of a particle moving along the x axis is defined by $x(t) = \frac{1}{3}t^3 - \frac{7}{2}t^2 + 12t + 2$

a. What is the initial position of the particle?

$x(0) = 0 - 0 + 0 + 2 = 2$

b. At what time(s) is the particle stopped?

$v(t) = 0 \qquad v(t) = x'(t) = t^2 - 7t + 12$

$t^2 - 7t + 12 = 0$

$(t-3)(t-4) = 0$

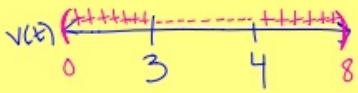
$t = 3 \quad t = 4$

c. On what interval(s) is the particle moving to the left?

$v(t) < 0$

$t^2 - 7t + 12 < 0$

$(t-3)(t-4) < 0$



moving to the left on $3 < t < 4$ because $v(t) < 0$

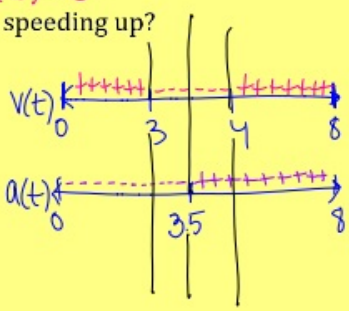
d. On what interval(s) is the particle speeding up?

$a(t) = v'(t) = 2t - 7$

$2t - 7 = 0$

$2t = 7$

$t = \frac{7}{2} = 3.5$

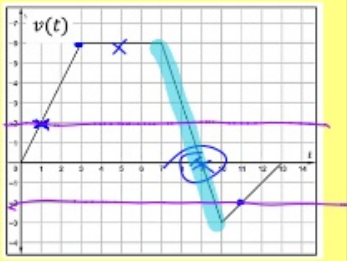


Speeding up on:

$3 < t < 3.5$ because $v(t) < 0$ and $a(t) < 0$

$4 < t < 8$ because $v(t) > 0$ and $a(t) > 0$

Example 2: The graph at the right models the velocity of a bug on the interval $0 \le t \le 13$.



a. Find $v(3)$ and $v(11)$

$V(3) = 6 \quad V(11) = -2$

b. Find $a(1)$, $a(5)$ and $a(9)$

$a(1) = v'(1) = 2$

$a(5) = v'(5) = 0$

$a(9) = v'(9) = -3$

c. At what time does the bug turn around? Give a reason for your answer.

The bug turns around at $t = 9$ because $v(t)$ changes from positive to negative at $t = 9$

d. On what interval does the bug have a negative acceleration?

$a(t) < 0 \longrightarrow v'(t) < 0$

on $7 < t < 10$ because $a(t) = v'(t) < 0$.

e. How many times is the speed of the bug equal to 2?

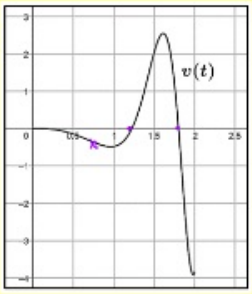
when $|v(t)| = 2$

$V(t) = 2$ or $V(t) = -2$

4 times

$(t = 1,\ t = 8.3,\ t = 9.7,\ t = 11)$

Example 3: (Calculator Active) For $0 \le t \le 2$, a bug is moving along the x-axis with $v(t) = t^2 \sin(t^3 - 1.5t)$



a. Find $a(0.7)$. Is the bug speeding up or slowing down at $t = 0.7$? Explain your reasoning.

$a(0.7) = v'(0.7) = -0.921$

$v(0.7) < 0$

Speeding up because $a(0.7) < 0$ and $v(0.7) < 0$

b. Find any value(s) of t in the interval $1 \le t \le 2$ when the bug is at rest.

when $v(t) = 0$: at $t = 1.225$ and $t = 1.801$

c. Find the open interval(s) on $0 < t < 2$ when the particle is moving to the right.

when $v(t) > 0$: on $1.225 < t < 1.801$

d. Find the acceleration of the bug when $v(t) = -1$.

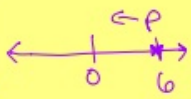
$v(t) = -1$ when $t = 1.8368729\ldots$ stored in Calc.

let $c = 1.8368729\ldots$

$a(c) = v'(c) = -28.874$

Example 4: For $0 \le t \le 8$ seconds, a particle P moves along the x-axis. The position, velocity and acceleration of particle P are differentiable and modeled by the functions x_P, v_P, and a_P respectively. Selected values for $x_P(t)$, $v_P(t)$, and $a_P(t)$ are shown in the table above, where $x_P(t)$ is measured in meters.

t (sec)	1	3	4	7
$x_P(t)$ (in meters)	2	8	6	5
$v_P(t)$	5	2	-1	-4
$a_P(t)$	-2	0	2	-3



a. Is particle P moving towards or away from the origin at time $t = 4$? Give a reason for your answer.

The particle is moving toward the origin because the position $x_P(4) = 6$, which is to the right of the origin, and $v_P(4) = -1$ means the particle is moving to the left, toward the origin.

b. Is particle P speeding up or slowing down at time $t = 7$? Give a reason for your answer.

The particle is speeding up because $v_P(7) = -4$ and $a_P(7) = -3$ are both negative (they have the same sign).

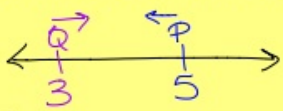
c. Use the data in the table to approximate $v_P(2)$. Indicate units of measure with your answer.

$$V_P(2) = X_P'(2) = \frac{X_P(3) - X_P(1)}{3-1} = \frac{8-2}{2} = \frac{6}{2} = 3 \text{ meters/sec}$$

d. Is there a time c, where $1 < c < 7$, such that $v_P(t) = 0$? Justify your answer.

Yes, by the IVT. The interval $(3, 4)$ is part of the interval $(1, 7)$. $v_P(t)$ is differentiable and therefore, continuous. The IVT guarantees since $v_P(4) = -1 < 0 < v_P(3) = 2$, then there is a number $t = c$ where c is between $t = 3$ and $t = 4$ for which $v_P(c) = 0$.

e. For $5 \le t \le 8$, a second particle Q moves along the same axis. The position of particle Q is modeled by $x_Q(t) = 12 - (t - 10)^2$. At time $t = 7$, are particles P and Q moving toward or away from each other? Explain your reasoning.



$X_P(7) = 5$

$V_P(7) = -4$ moving left

$X_Q(7) = 12 - (7-10)^2 = 12 - 9 = 3$

$V_Q(7) = -2(7-10) = -2(-3) = 6$ moving Right

$V_Q(t) = -2(t-10)$

They are moving toward each other because the Position of Particle P is $X_P(7) = 5$, and moving left and Particle Q is at $X_Q(7) = 3$ and moving right.